

EXCEL – CLASS 1

Introduction to Data, Excel & Course Syllabus

1. What is Data?

Data means raw facts, values, observations, or information that can be collected and processed. Examples include names, ages, marks, product prices, sales quantities, dates, employee details, and customer information.

Example: Piyush, 22, B.Tech CSE, 7.10 CGPA — these are individual pieces of data about a student.

2. Why Do We Use Data?

- To store information in an organized way.
- To calculate and compare values.
- To identify patterns and trends.
- To create reports and summaries.
- To support business and personal decision-making.
- To analyze large amounts of information more efficiently.

3. What is Microsoft Excel?

Microsoft Excel is a spreadsheet application used to store, organize, calculate, analyze, and visualize data. Excel stores information in rows and columns, where their intersection is called a cell.

4. Why Do We Use Excel?

- Data entry and organized data storage.
- Calculations using formulas and functions.
- Sorting and filtering data.
- Data cleaning and transformation.
- Charts and visual reporting.
- PivotTables and PivotCharts for summarizing data.
- Conditional formatting for highlighting important information.
- Data analysis and business reporting.
- Automation using VBA and Macros in advanced Excel.

5. Basic Excel Concepts

- **Workbook:** An Excel file that can contain multiple worksheets.
- **Worksheet:** A sheet containing rows and columns.
- **Row:** A horizontal line of cells, identified by numbers.
- **Column:** A vertical line of cells, identified by letters.

- **Cell:** The intersection of a row and a column, such as A1.
- **Range:** A group of cells, such as A1:A10.
- **Formula:** An expression used to perform a calculation, usually beginning with =.
- **Function:** A predefined Excel operation that can accept arguments and return a result, such as SUM, IF, LEFT, RIGHT, and LEN.

6. Example: Formula vs Function

Formula: =A1+B1 — adds the values in A1 and B1.

Function: =SUM(A1:A8) — uses the SUM function to add values from A1 through A8.

Text function examples: =UPPER(A1), =LOWER(A1), =LEFT(A1,3), =RIGHT(A1,3), =LEN(A1).

EXCEL & ADVANCED EXCEL – COURSE SYLLABUS

Beginner to Advanced • Practical, Formula, Analysis & Automation Skills

Basic Excel

- Excel tutorial
- Text to Columns
- Concatenate
- The CONCATENATE Function
- The RIGHT Function with Concatenation
- Absolute Cell References
- Data Validation
- Time and Date Calculations
- Conditional Formatting
- Exploring Styles and Clearing Formatting
- Using Conditional Formatting to Hide Cells
- Using the IF Function
- Changing the “Value if false” Condition to Text

Pivot Tables & PivotCharts

- Pivot Tables
- Creating a Pivot Table
- Specifying PivotTable Data
- Changing a PivotTable's Calculation
- Filtering and Sorting a PivotTable
- Creating a PivotChart
- Grouping Items
- Updating a PivotTable
- Formatting a PivotTable
- Using Slicers

Charts

- Creating a Simple Chart
- Charting Non-Adjacent Cells
- Creating a Chart Using the Chart Wizard
- Modifying Charts

- Moving an Embedded Chart
- Sizing an Embedded Chart
- Changing the Chart Type
- Chart Types
- Changing the Way Data is Displayed
- Moving the Legend
- Formatting Charts
- Adding Chart Items
- Formatting All Text
- Formatting and Aligning Numbers
- Formatting the Plot Area
- Formatting Data Markers

Pie Charts

- Creating a Pie Chart
- Moving the Pie Chart to its Own Sheet
- Adding Data Labels
- Exploding a Slice of a Pie Chart

Data Analysis & Range Names

- Data Analysis – Overview
- Types of Data Analysis
- Data Analysis Process
- Working with Range Names
- Copying Names using Formula Autocomplete
- Range Name Syntax Rules
- Creating Range Names
- Creating Names for Constants
- Managing Names
- Scope of a Name
- Editing Names
- Applying Names
- Using Names in a Formula
- Viewing Names in a Workbook
- Copying Formulas with Names

Tables & Data Management

- Difference between Tables and Ranges
- Create Table
- Table Name
- Managing Names in a Table
- Table Headers replacing Column Letters
- Propagation of a Formula in a Table
- Resize Table
- Remove Duplicates
- Convert to Range
- Table Style Options
- Table Styles

Text Functions & Data Cleaning

- Cleaning Data with Text Functions
- Removing Unwanted Characters from Text
- Extracting Data Values from Text
- Formatting Data with Text Functions

Advanced Excel

- Date Formats
- Conditional Formatting
- Sorting
- Filtering
- Lookup Functions
- Pivoting
- 50+ dynamic Excel formulas and functions
- Excel formatting
- VBA
- Macros
- Hands-on demonstrations and practical exercises

Course Goal

The course is designed to take a learner from Excel fundamentals to advanced data analysis and automation. By the end, the learner should be able to organize data, use formulas and functions, clean and analyze datasets, create reports and dashboards with PivotTables and charts, and understand VBA/Macros for automation.

